

Comprehensive Edumeup UAT & QA Checklist

This document is an exhaustive, line-by-line checklist derived from the 13-page Edumeup Software Requirements Specification (SRS). It covers Functional Requirements, Non-Functional Requirements, Integrations, and Architecture.

Testing Key:

- **Pass:** Feature works exactly as expected and documented in the SRS.
- **Fail:** Feature is broken, incomplete, or does not behave as specified.
- **Pending:** Feature not yet tested or currently being fixed by the developer.

1. Authentication (SRS 4.1, 4.19, 8, 9)

ID	Requirement / Test Case	Status	Notes / Dev Feedback
1.1	User Registration: Users can successfully create an account.	[]	
1.2	Account Creation Flow: On sign up, a local account AND a Moodle user are created.	[]	
1.3	Email Verification: System triggers an email verification step after registration.	[]	Hostinger SMTP setup is required
1.4	Login/Logout: Users can securely log in and log out.	[]	
1.5	Password Reset: "Forgot Password" sends a reset link; users can reset securely.	[]	
1.6	Change Password: Logged-in users can update their password.	[]	
1.7	Session Management: Sessions are tracked; users are logged out after timeout.	[]	
1.8	JWT Authentication: Backend relies on JSON Web Tokens	[]	Backend relies on Secure HTTP-Only Cookies + Server-Side Sessions (backed by PostgreSQL) for API authentication. This approach is more secure than client-side JWT storage because it is immune to XSS attacks, allows instant session revocation, and integrates seamlessly with our existing PostgreSQL database.

ID	Requirement / Test Case	Status	Notes / Dev Feedback
	(JWT) for API auth.		
1.9	SSO Integration (4.19): Seamless course access via auto-authentication into Moodle using secure tokens.	[]	

2. User Profile & Role Management (SRS 4.2, 4.3)

ID	Requirement / Test Case	Status	Notes / Dev Feedback
2.1	Update Details: Users can update their personal details.	[]	
2.2	Profile Image: Users can upload a profile image (saved in AWS S3).	[]	
2.3	Contact Info: Users can manage their contact information.	[]	
2.4	Account Activity: Users can view their recent account activity.	[]	
2.5	RBAC (Student): Students have restricted access to learning and purchase areas.	[]	
2.6	RBAC (Parent): Parents can access child performance reports, blocked from admin areas.	[]	
2.7	RBAC (School): Organizations can access bulk purchase and user assignment pages.	[]	
2.8	RBAC (Admin): Admins have full system control.	[]	

3. Course Catalog & Search (SRS 4.4, 4.22)

ID	Requirement / Test Case	Status	Notes / Dev Feedback
3.1	Catalog Display: Courses fetched from Moodle display correctly on the frontend.	[]	
3.2	Full-Text Search: Users can search courses using a full-text search engine.	[]	
3.3	Filters: Users can filter courses by relevant categories.	[]	
3.4	Sorting & Pagination: Course results support sorting (e.g., price, date) and pagination.	[]	
3.5	Course Details: Clicking a course displays its full description, pricing, and details.	[]	
3.6	Add to Cart: Users can add the course to their shopping cart from the details page.	[]	

4. Cart, Orders & Payments (SRS 4.7, 4.8, 4.9, 8)

ID	Requirement / Test Case	Status	Notes / Dev Feedback
4.1	Cart Management: Users can add/remove items, apply coupons, calculate totals.	[]	
4.2	Checkout Flow: Users checkout, redirect to PayFast Pakistan , and pay securely.	[]	
4.3	Webhooks: PayFast webhooks update the system upon successful/failed transaction.	[]	

ID	Requirement / Test Case	Status	Notes / Dev Feedback
4.4	Payment Logs: System stores comprehensive logs of the payment transaction.	[]	
4.5	Partial Payments: System properly supports and logs partial payments.	[]	
4.6	Retries & Chargebacks: Retry mechanism for failed payments; handling for disputes.	[]	
4.7	Order History: Users can view their past orders and track payment statuses.	[]	
4.8	Invoices: Users can download an invoice for completed orders.	[]	

5. Course Enrollment (SRS 4.5)

ID	Requirement / Test Case	Status	Notes / Dev Feedback
5.1	Post-Payment Enrollment: System automatically enrolls user via Moodle after payment.	[]	
5.2	Enrollment Sync: Enrollment status matches between Custom Backend and Moodle.	[]	
5.3	Enrolled Courses Display: Courses successfully appear in the user's dashboard.	[]	

6. Dashboards (SRS 4.6, 4.10, 4.11, 4.12, 4.18)

6A. Student Dashboard

ID	Requirement / Test Case	Status	Notes / Dev Feedback
6A.1	Dashboard View: Displays enrolled courses, completion status, and grades.	[]	
6A.2	Activity Timeline: Student can see an activity timeline of their learning.	[]	
6A.3	Notifications: In-app notification bell/feed is visible and functioning.	[]	
6A.4	Certificates (4.18): Certificates sync from Moodle and are downloadable.	[]	

6B. Parent Dashboard

ID	Requirement / Test Case	Status	Notes / Dev Feedback
6B.1	Children Profiles: Parent can connect to and view their children's profiles.	[]	
6B.2	Progress Monitor: Parent can view child's grades and course progress.	[]	
6B.3	Alerts & Reports: Parent receives alerts and can download performance reports.	[]	

6C. School / Organization Dashboard

ID	Requirement / Test Case	Status	Notes / Dev Feedback
6C.1	Bulk Purchase: School can purchase multiple licenses for a	[]	

ID	Requirement / Test Case	Status	Notes / Dev Feedback
	course.		
6C.2	Seat Assignment: Schools can assign purchased licenses to specific students.	[]	
6C.3	CSV Upload: Schools can bulk-upload a list of students via CSV.	[]	
6C.4	Usage & Reports: School admin can track student usage and generate reports.	[]	

6D. Admin Management Dashboard (4.12, 4.13, 4.14)

ID	Requirement / Test Case	Status	Notes / Dev Feedback
6D.1	User Management: Admin can suspend accounts, reset passwords, assign roles.	[]	
6D.2	Activity Logs: Admin can view detailed activity logs across the platform.	[]	
6D.3	Course Sync: Admin can manually/automatically sync courses from Moodle.	[]	
6D.4	Course Pricing/Visibility: Admin can set pricing, categories, and visibility.	[]	
6D.5	Analytics: Admin can generate revenue, enrollment, progress, and usage metrics.	[]	

7. Forms, Support & Notifications (SRS 4.15, 4.16, 4.17)

ID	Requirement / Test Case	Status	Notes / Dev Feedback
7.1	Forms: Contact forms, teacher apps, and custom forms work and save to DB.	[]	
7.2	Form File Uploads: Form file uploads successfully store in AWS S3.	[]	
7.3	CSV Export: Admin can export form submissions as a CSV.	[]	
7.4	Support System: Users can submit tickets, track status, and get responses.	[]	
7.5	Notifications: Email & in-app notifications send for payments and course updates.	[]	

8. Background Processing & Error Handling (SRS 4.20, 4.21)

ID	Requirement / Test Case	Status	Notes / Dev Feedback
8.1	Queues: Email, notifications, payment verification, Moodle sync run in background.	[]	
8.2	API Failures: Moodle or PayFast API failures are handled gracefully without crashing.	[]	
8.3	Error UI: Users receive friendly error messages on failure, not raw stack traces.	[]	
8.4	Retry Mechanism: System utilizes retry mechanisms for transient API failures.	[]	
8.5	Dead Letter Queue:	[]	

ID	Requirement / Test Case	Status	Notes / Dev Feedback
	Failed background jobs route to a dead letter queue for review.		

9. Non-Functional & Architecture Requirements (SRS 3, 5, 6, 7, 10, 11, 12)

Note: QA team will need developer/DevOps assistance to verify the below points.

ID	Requirement / Test Case	Status	Notes / Dev Feedback
9.1	Tech Stack Validation: Frontend is React/Next.js; Backend is Node.js; DB is PostgreSQL.	[]	
9.2	Database Schema: Tables exist (users, roles, profiles, courses_cache, orders, payments, licenses, license_users, notifications, forms, form_fields, form_submissions, activity_logs).	[]	
9.3	Performance (5.1): Pages load in under 3 seconds.	[]	
9.4	Caching (5.6): Redis caching layer, CDN, and Session caches are implemented.	[]	
9.5	Security (5.2, 9): Rate limiting, data encryption, secure cookies, input validation, audit logs are active.	[]	
9.6	Moodle API Integration (7): Integrates for Course retrieval, Enrollment, Tracking, Grades. Strict failure/rate	[]	

ID	Requirement / Test Case	Status	Notes / Dev Feedback
	limiting handling is in place.		
9.7	Logging (10): Login attempts, payment transactions, user activity, and errors are logged.	[]	
9.8	CI/CD & Environments (11): Automated testing/deployment pipelines; dev/staging/prod are separated.	[]	
9.9	Infrastructure (12): Deployed with Cloud hosting, Managed database, CDN, and Backups.	[]	
9.10	Usability (5.5): UI is highly responsive on mobile devices and accessible.	[]	